

ASHG
ANNUAL MEETING
MONTRÉAL
OCTOBER 20-24, 2026

Modeling Choices in Omics

Implications for Health Disparities in Human Genetics

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Financial Disclosure

We do not have any relationships to report within the last 24 months with ACCME defined ineligible companies.

Unlabeled/Investigational Uses

We will not be discussing unlabeled/investigational uses of medical devices or pharmaceuticals during this presentation.

Join the platform

ASHG 2026 · MODELING CHOICES IN OMICS

heart-gen.github.io/
modeling-choices-omics-ashg2026

No account, no install.

Opening poll:

How does your team usually evaluate ancestry-related heterogeneity?

Two-hour agenda

AGENDA

Segment	Time
Opening and live poll	15 min
Framing and prediction practice	15 min
Interactive exercises I	30 min
Interactive exercises II	30 min
Synthesis	15 min
Wrap-up and Q&A	15 min

How the modules work

EVERY MODULE RUNS THE SAME WAY

1

Commit

Answer before any chart appears

2

Room split

See how the room divided

3

Explore

Move every control

4

Debrief

What makes another answer right?

Pooled vs Stratified Analysis

MODULE 1 · COMMIT FIRST

Raise the European-ancestry share of the pooled sample from 35% to 95%. What happens to the pooled effect estimate?

A. It grows toward the strong European-ancestry effect

B. It shrinks toward the weak African-ancestry effect

C. It stays put; only the interval changes

D. It stops being significant

Pooled vs Stratified Analysis: the answer

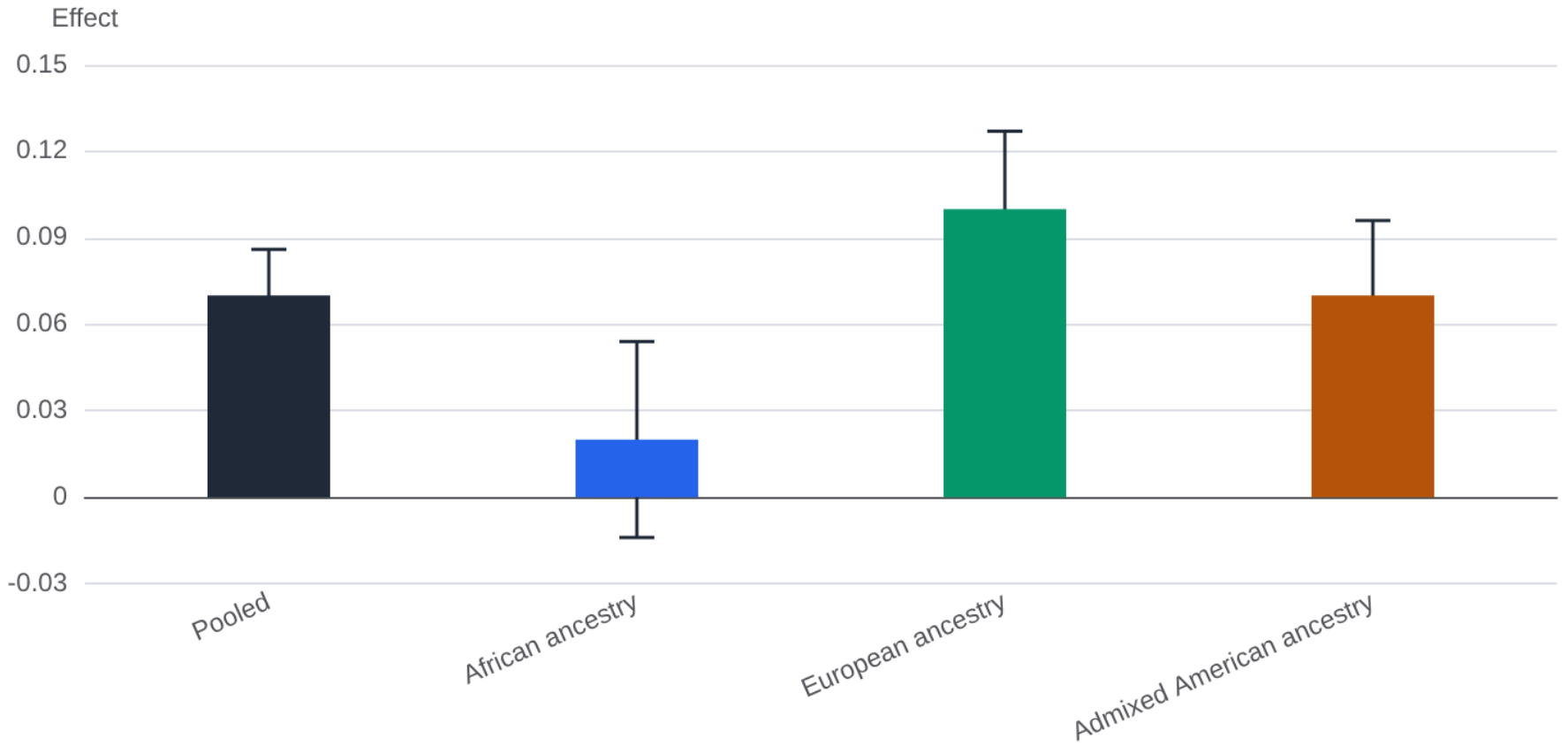
MODULE 1 · REVEAL

Answer

It grows toward the strong European-ancestry effect

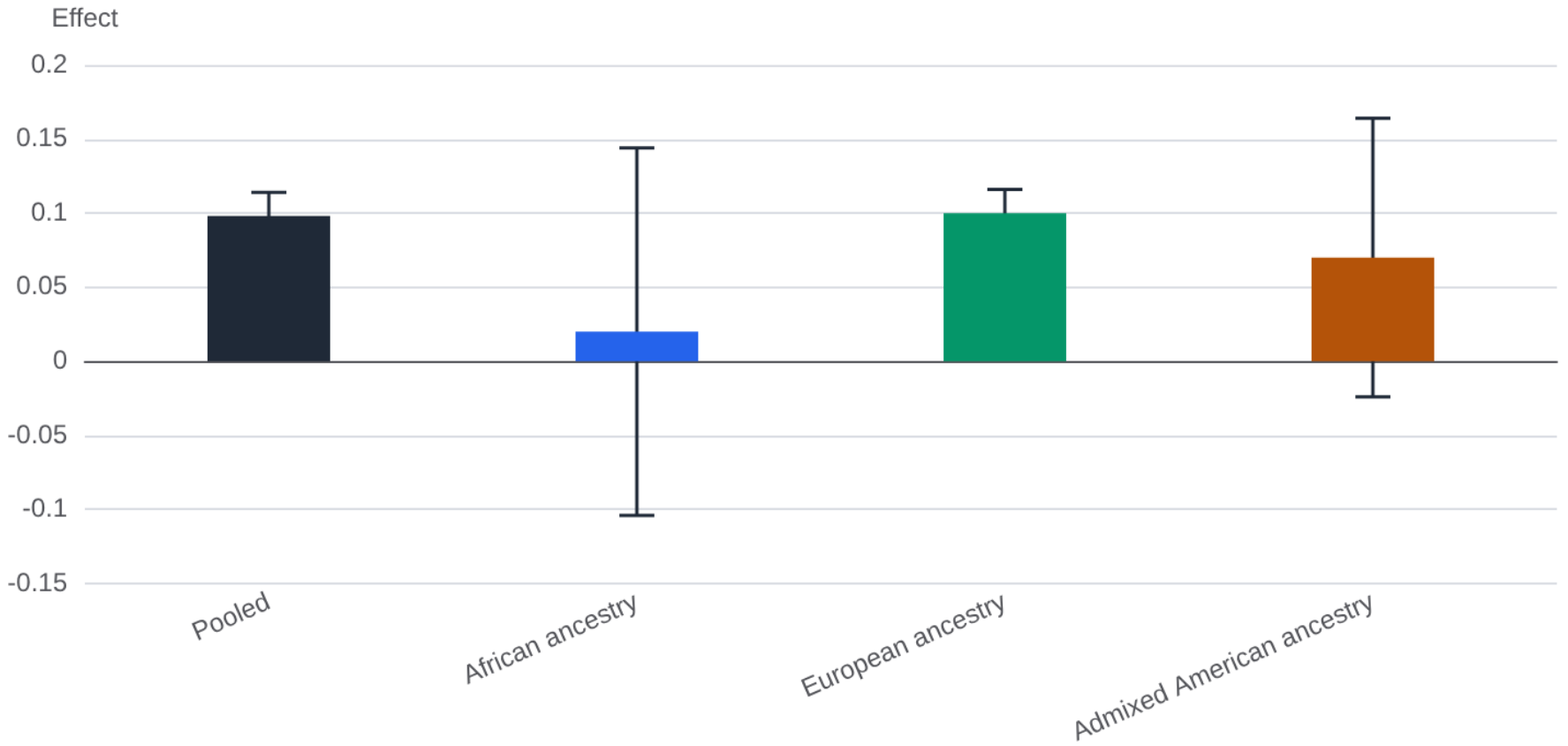
Module 1: Effect estimates at 35% European ancestry

MODULE 1 · REVEAL



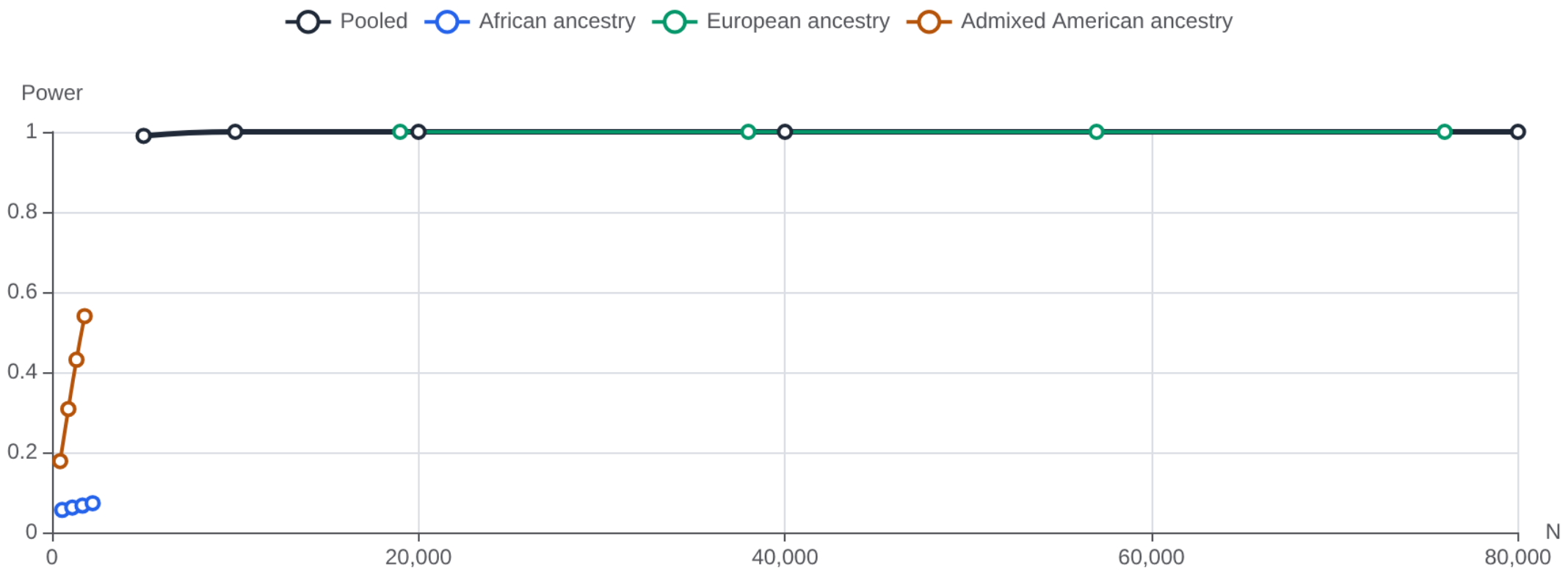
Module 1: Effect estimates at 95% European ancestry

MODULE 1 · REVEAL



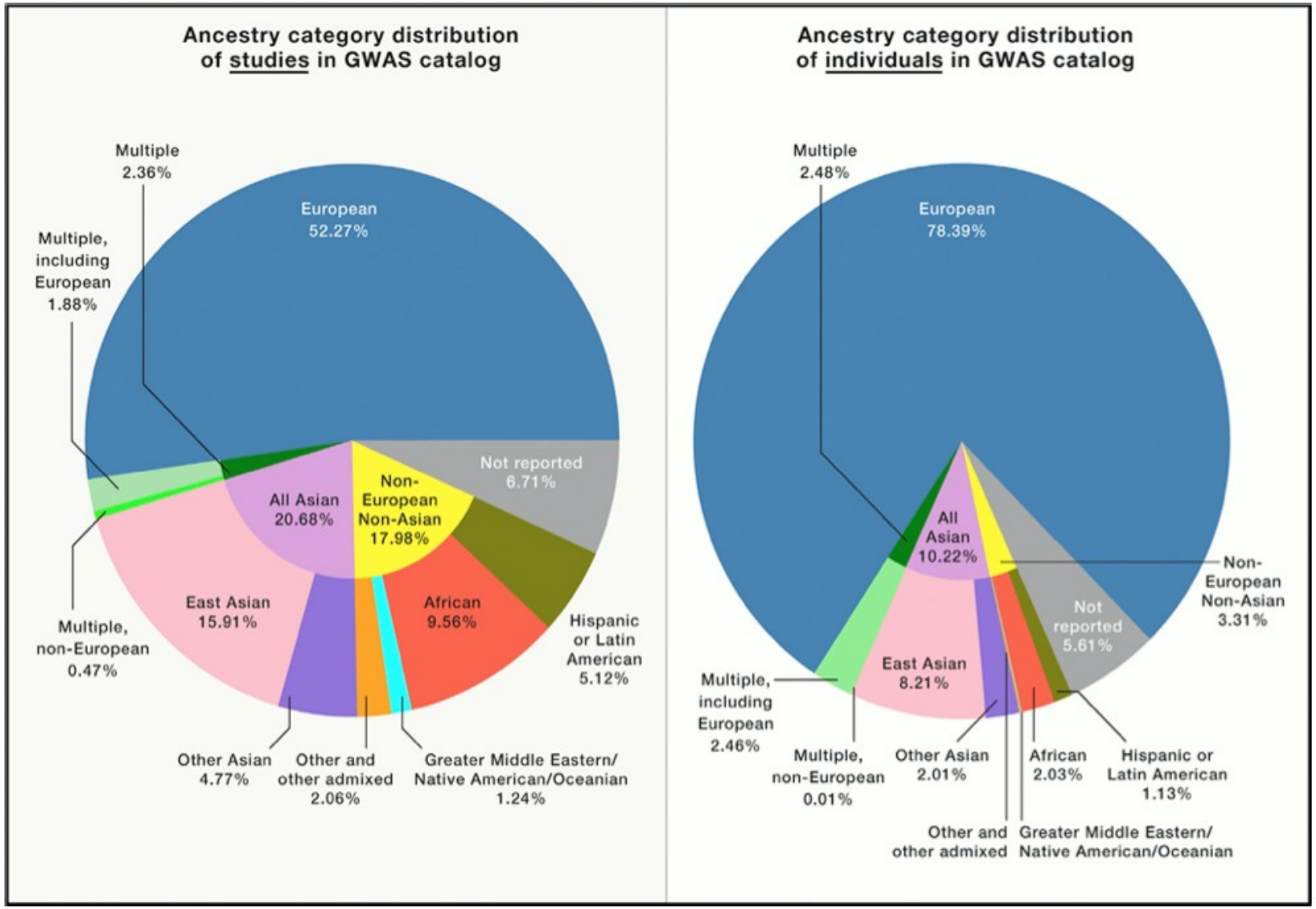
Module 1: Power curve at 95% European ancestry

MODULE 1 · REVEAL



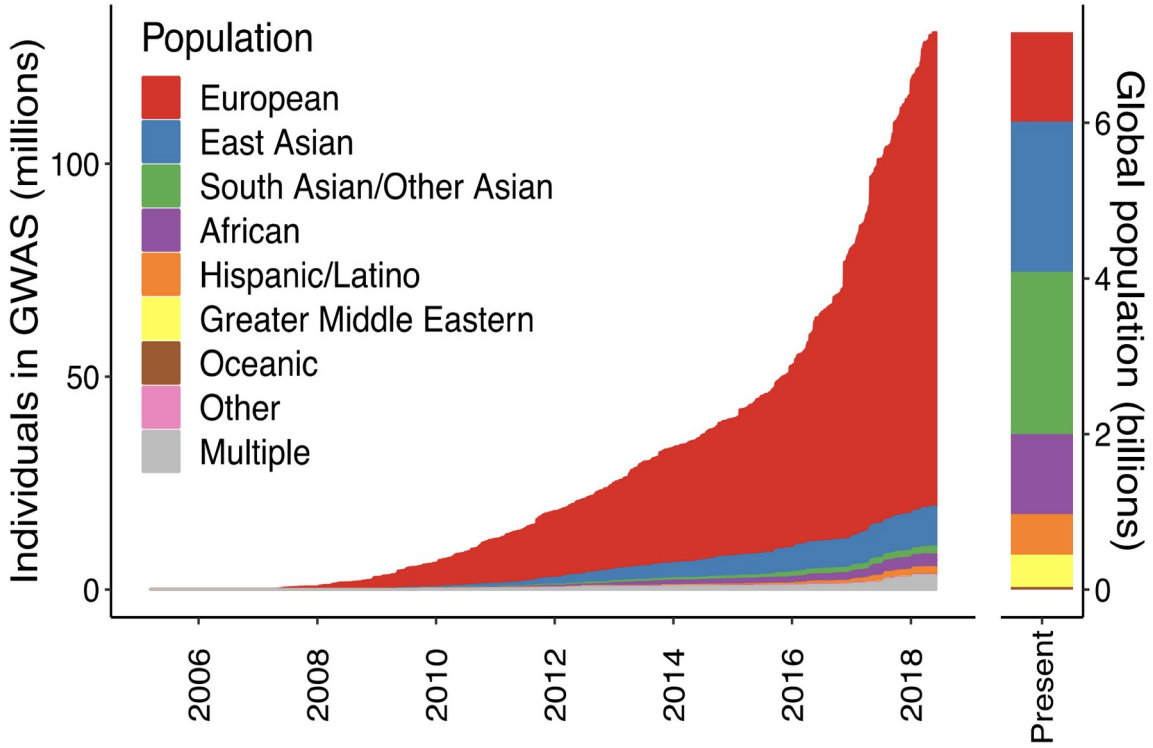
Genomics today

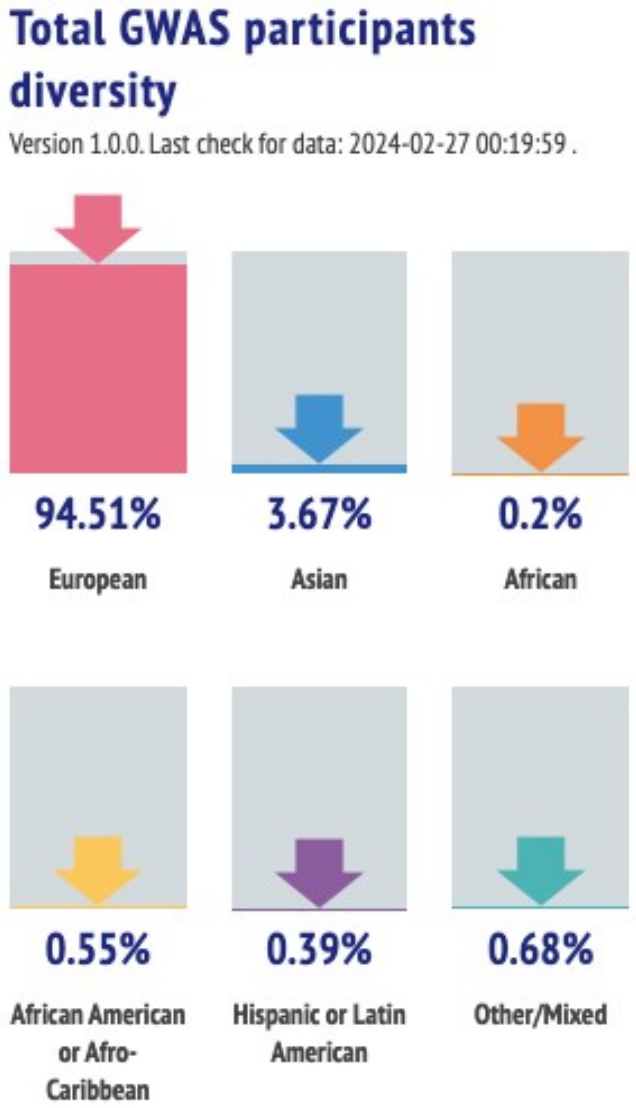
More studies are STILL in predominantly European-descent populations.



Sirugo et al, *Cell*, 2019

Genomics today



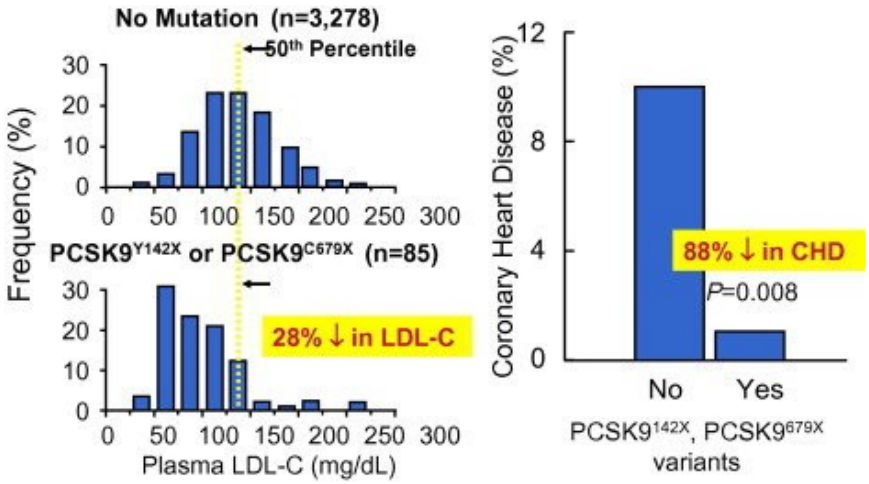


More studies are STILL in predominantly European-descent populations.

Sirugo et al, *Cell*, 2019

Apolipoprotein E and Alzheimer's disease pathology in African American older adults

Sarah K. Royse [a b], Beth E. Snitz [c], Ashley V. Hill [a], Alexandria C. Reese [b], Rebecca E. Roush [c], M. Ilyas Kamboh [a d e], Marnie Bertolet [a f], Anum Saeed [g], Brian J. Lopresti [b], Victor L. Villemagne [d], Oscar L. Lopez [c d], Steven E. Reis [g], James T. Becker [c d h], Ann D. Cohen [d]



Sirugo et al. The Missing Diversity in Human Genetic Studies. Cell 2019
Fruchart, International Journal of Diabetes Mellitus, 2010.

The apolipoprotein-E4 (*APOE*4*) and apolipoprotein-E2 (*APOE**2) alleles are more common in African American versus non-Hispanic white populations, but relationships of both alleles with Alzheimer's disease (AD) pathology among African American individuals are unclear. We measured *APOE* allele and β-amyloid (Aβ) and tau using blood samples and positron emission tomography (PET) images, respectively. Individual regression models tested associations of each *APOE* allele with Aβ or tau PET overall, stratified by racialized group, and with a racialized group interaction. We included 358 older adults (42% African American) with Aβ PET, 134 (29% African American) of whom had tau PET. *APOE*4* was associated with higher Aβ in non-Hispanic white ($P < 0.0001$), but not African American ($P = 0.64$) participants; racialized group modified the association between *APOE*4* and Aβ ($P < 0.0001$). There were no other racialized group differences. These results suggest that the association of *APOE*4* and Aβ differs between African American and non-Hispanic white populations. Other drivers of AD pathology in African American populations should be identified as potential therapeutic targets.

Within-Ancestry Testing vs Meta-Analysis

MODULE 2 · COMMIT FIRST

Four ancestry groups are meta-analysed. Push heterogeneity to 100%, where the East Asian effect is about 3 times the others'. What happens to the fixed-effect summary?

A. It stays narrow and significant

B. Its interval widens like the random-effects one

C. It stops being significant

D. It flips sign

Within-Ancestry Testing vs Meta-Analysis: the answer

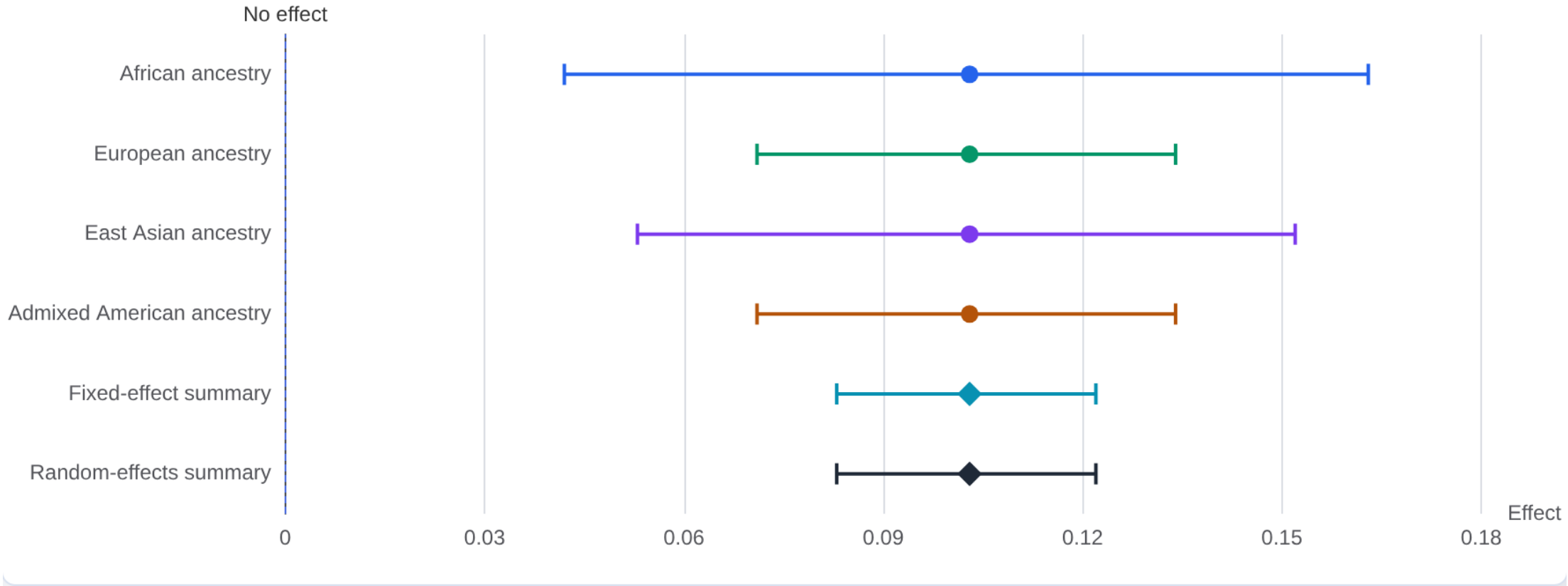
MODULE 2 · REVEAL

Answer

It stays narrow and significant

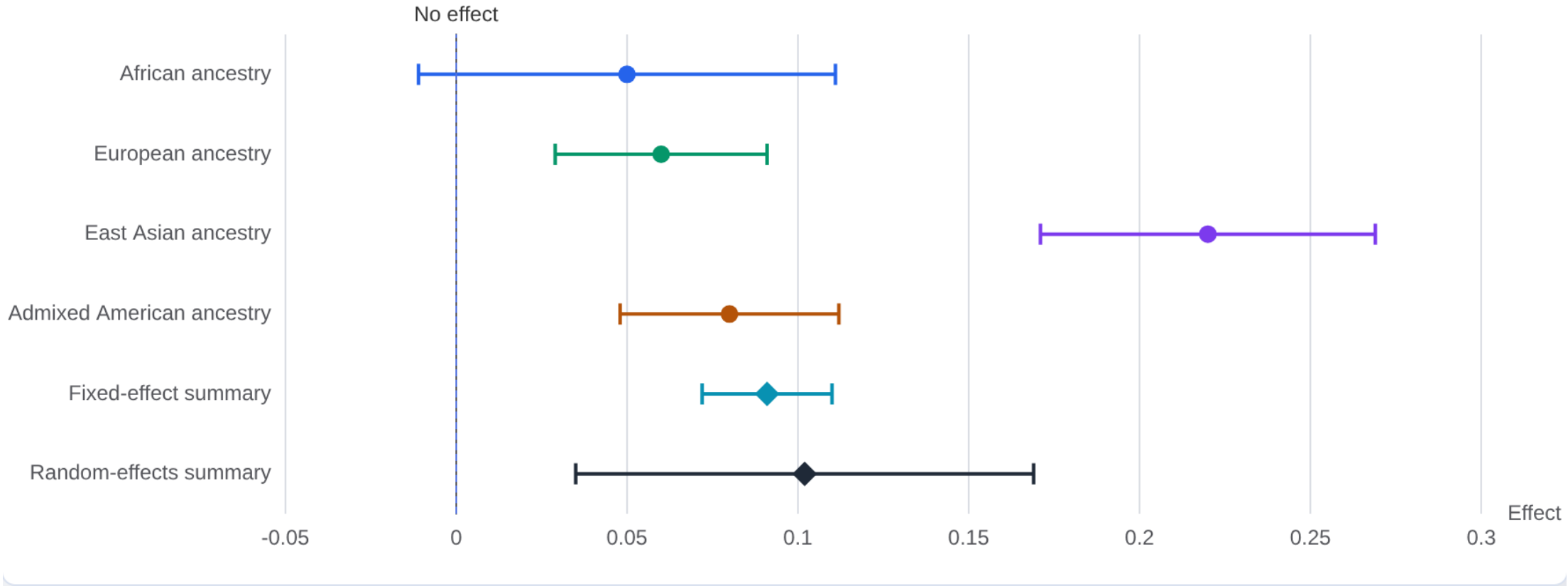
Module 2: Forest plot at 0% heterogeneity

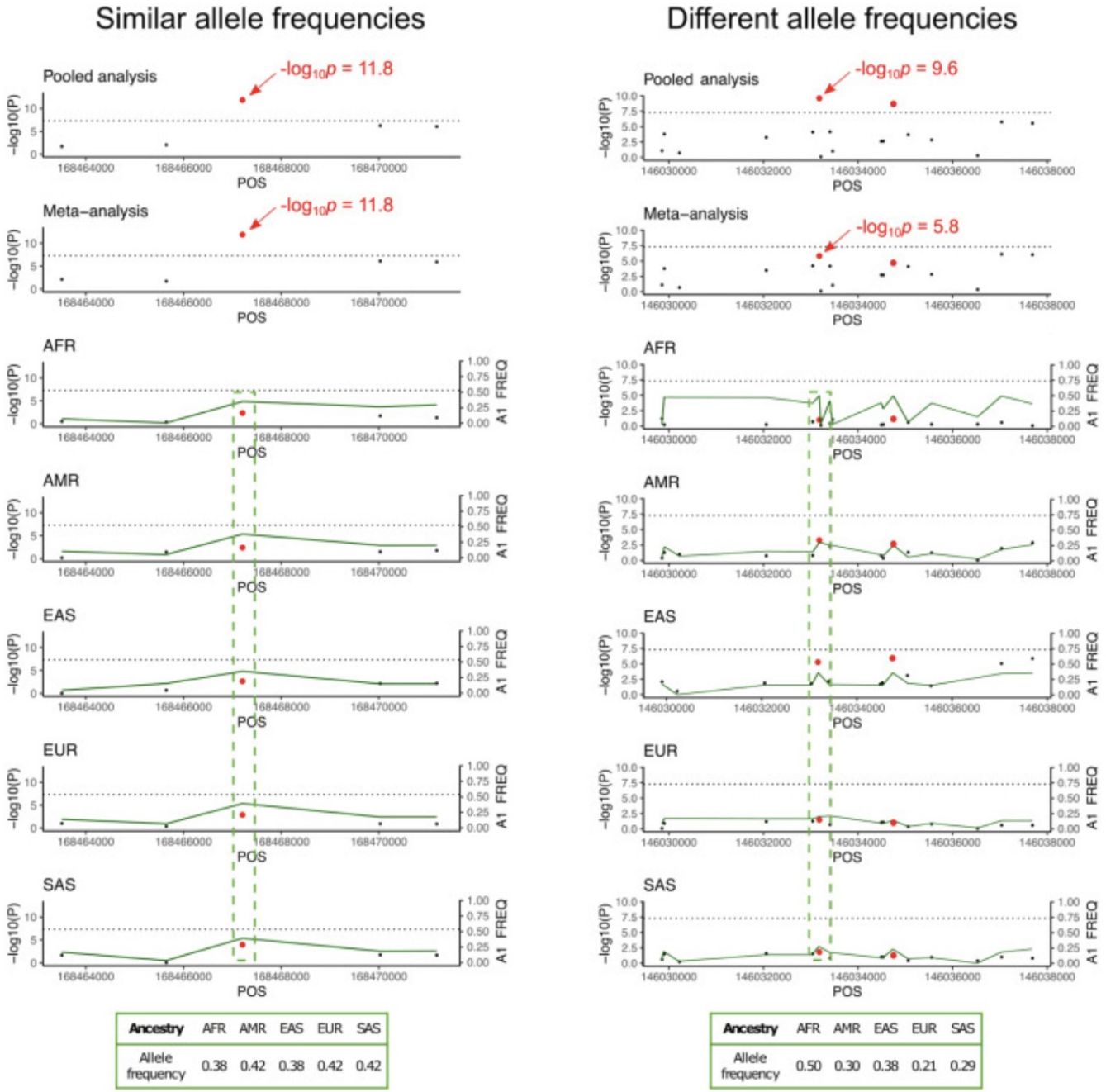
MODULE 2 · REVEAL



Module 2: Forest plot at 100% heterogeneity

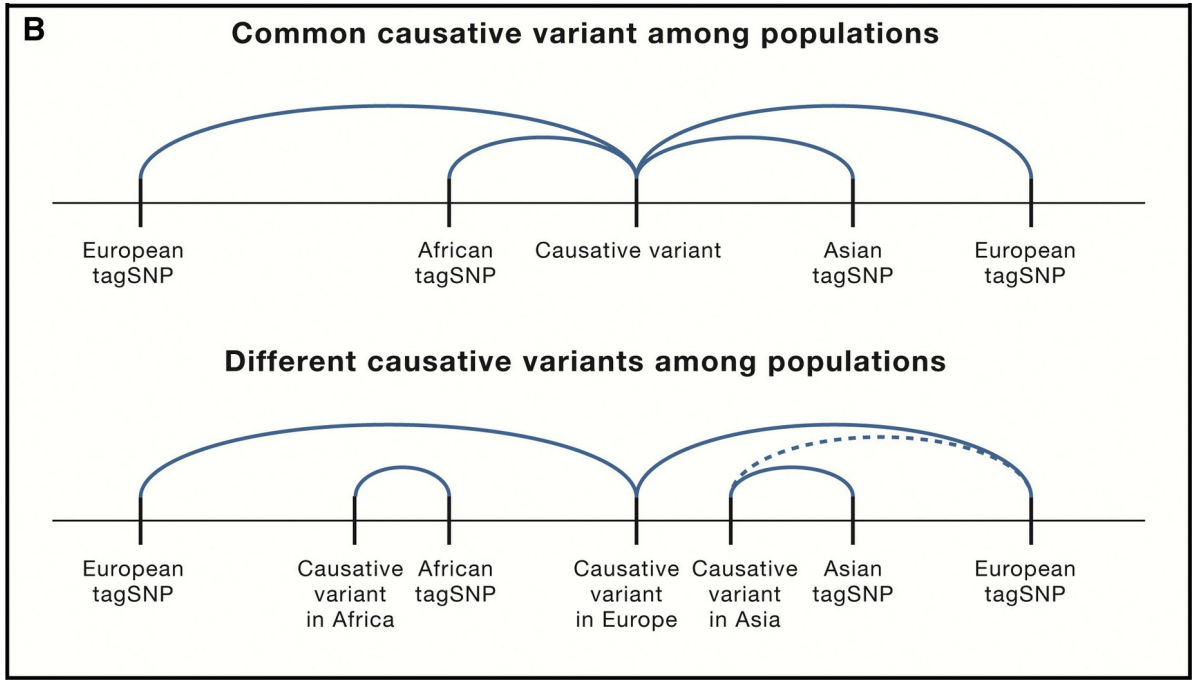
MODULE 2 · REVEAL





Dias et al, AJHG, 112(10), 2 October 2025, Pages 2493-2508

Opposite effects by ancestry: which model detects them?



"Even when diverse populations are studied, specific gene effects in these populations may not be evident, **as often diverse populations are studied only as part of large meta-analyses that estimate associations from combined data**. The result of this analytical strategy is to identify variants that have mostly similar effects across populations, but it can reduce the ability to detect population-specific genetic risk factors."

Cell. 2019 Mar 21;177(1):26–31.

Global vs Local Ancestry Modeling

MODULE 3 · COMMIT FIRST

The tag SNP has opposite effects on African- and European-ancestry haplotypes (+0.12 and −0.08) in a 15,000-person admixed study. Which is the simplest analysis that detects it at p < 0.05?

A. Global ancestry PCs

B. Adding local ancestry as a covariate

C. Local ancestry interaction

D. None at this sample size

Global vs Local Ancestry Modeling: the answer

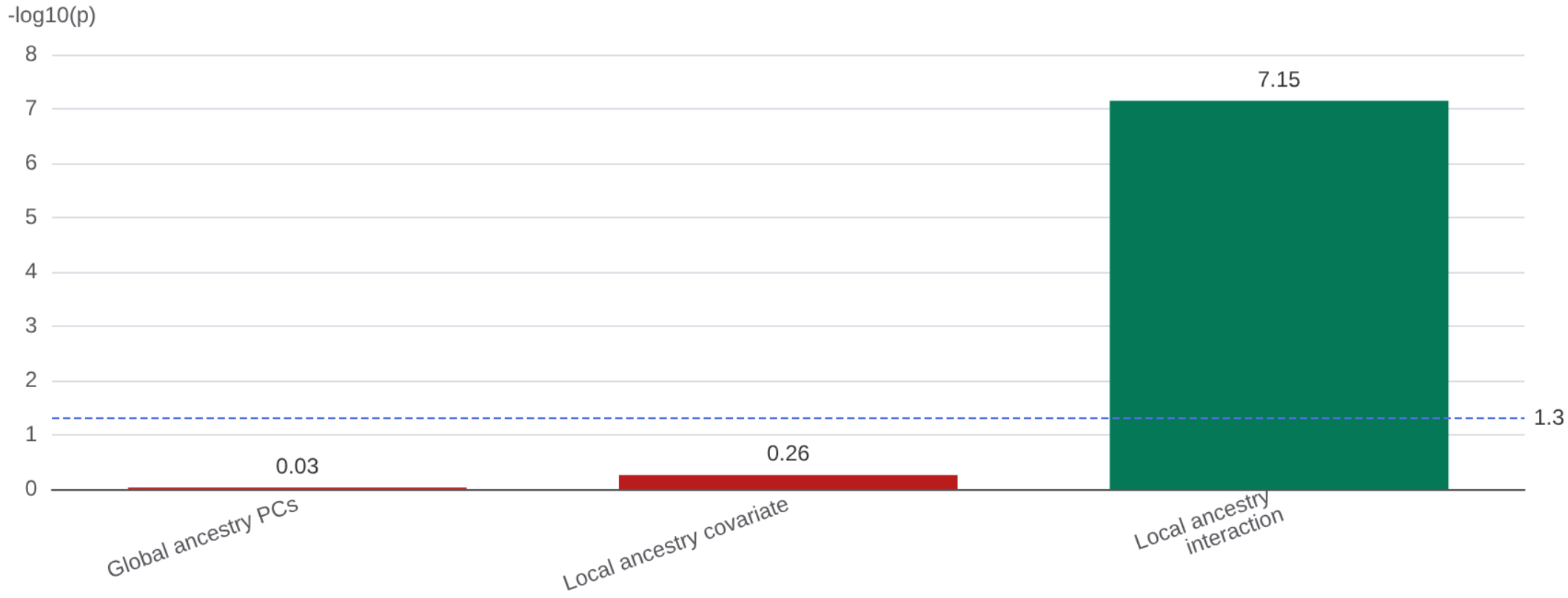
MODULE 3 · REVEAL

Answer

Local ancestry interaction

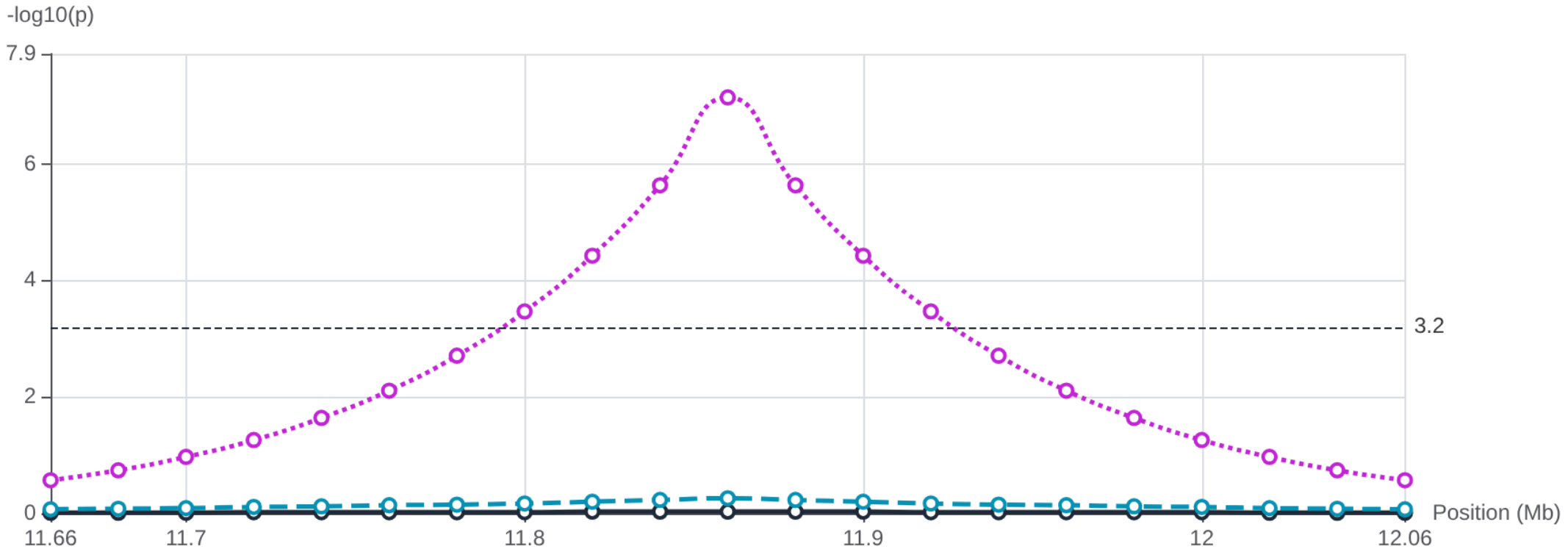
Module 3: Does each model detect the locus?

MODULE 3 · REVEAL

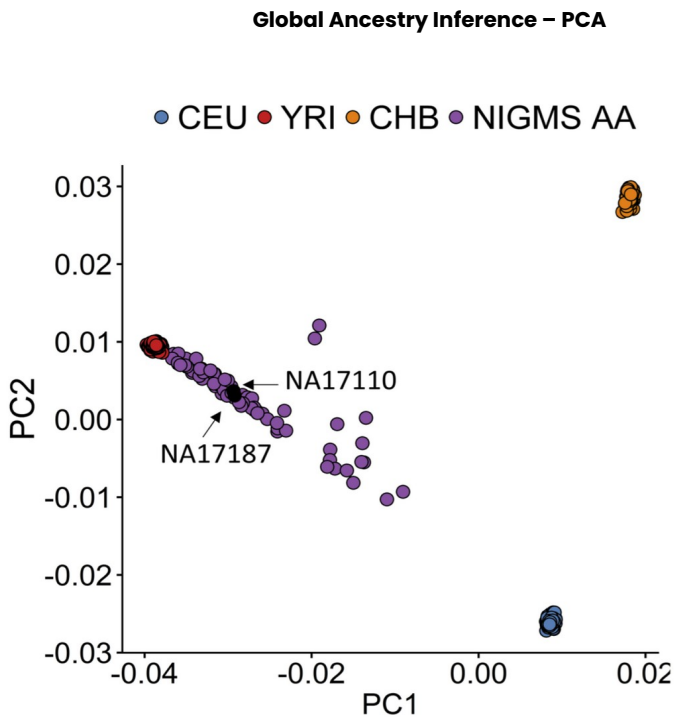


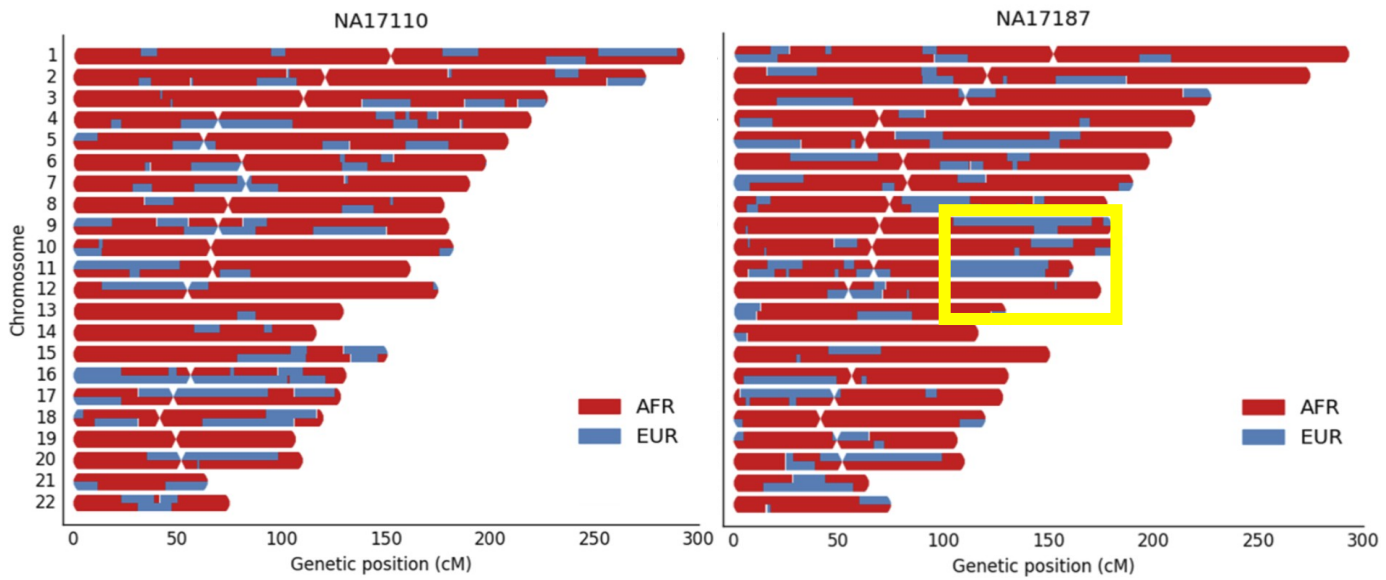
Module 3: Regional signal under each model

MODULE 3 · REVEAL



Local Ancestry Inference





Zhang *et al.*, Clinical Pharmacology & Therapeutics, 2019

Article | Published: 18 January 2021

Tractor uses local ancestry to enable the inclusion of admixed individuals in GWAS and to boost power

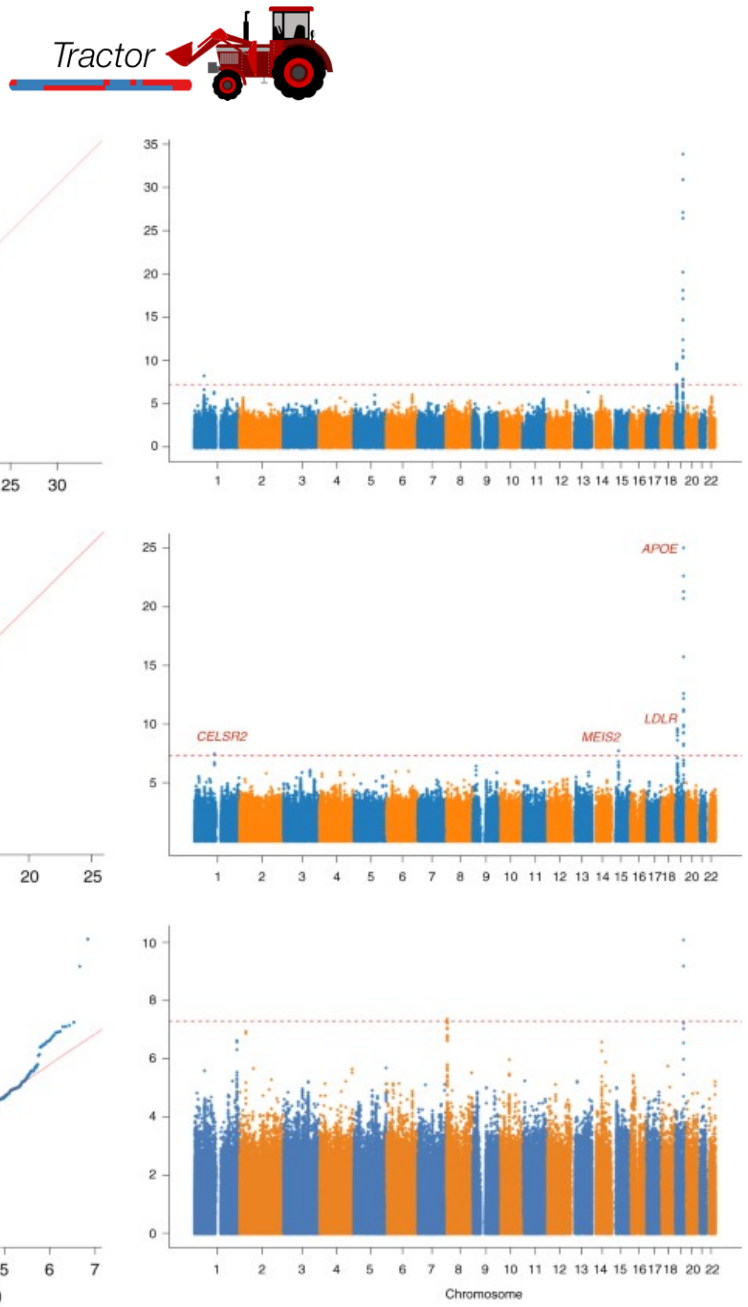
Elizabeth G. Atkinson ✉, Adam X. Maihofer, Masahiro Kanai, Alicia R. Martin, Konrad J. Karczewski, Marcos L. Santoro, Jacob C. Ulirsch, Yoichiro Kamatani, Yukinori Okada, Hilary K. Finucane, Karestan C. Koenen, Caroline M. Nievergelt, Mark J. Daly & Benjamin M. Neale

Nature Genetics **53**, 195–204 (2021) | Cite this article

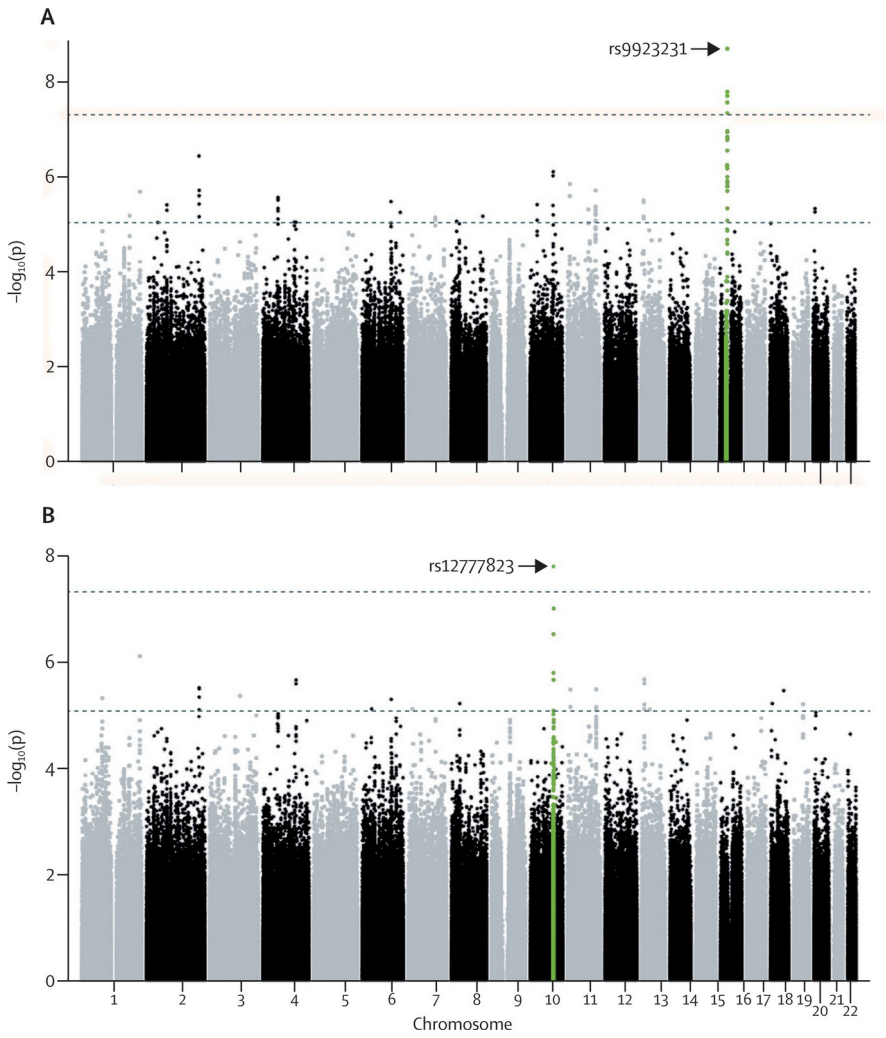
21k Accesses | **108** Citations | **139** Altmetric | Metrics

Matters Arising to this article was published on 25 November 2021

- Allows for the interrogation of each ancestry segment separately.
- These separate ancestry-specific GWAS can then be meta-analyzed.
- Increased the power to detect novel signals that differ in allele frequencies between populations.

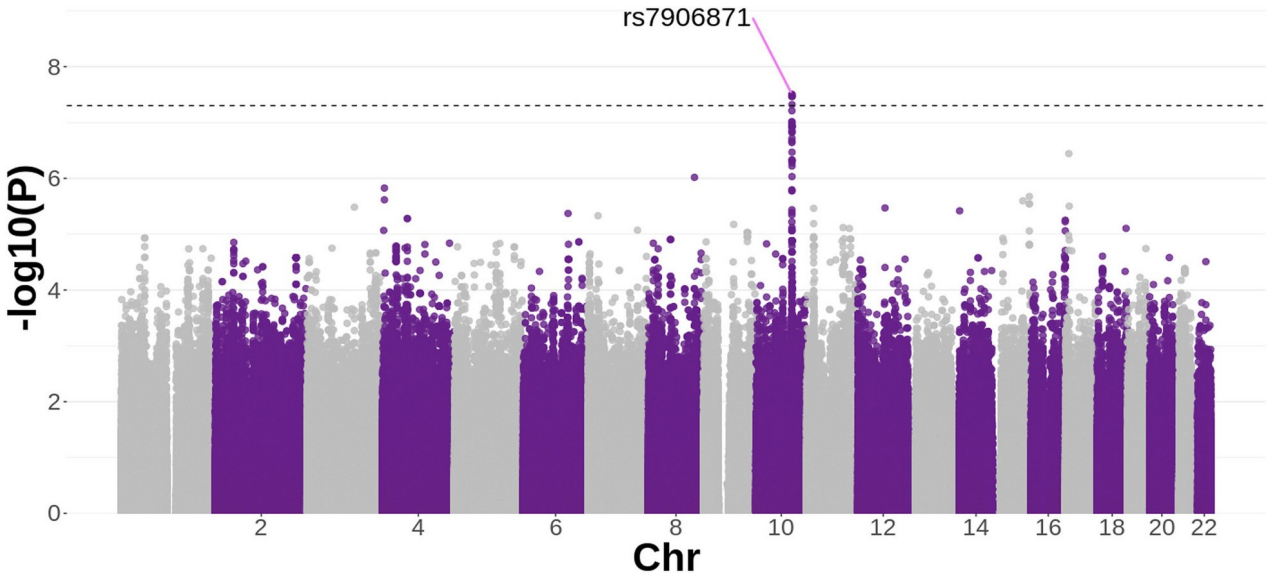


Traditional GWAS



Lancet. 2013 Aug31;382(9894):790-6

TRACTOR GWAS



rs12777823 achieved *p-values* of 2.2E-6 in the African tract and 8.3E-6 in the meta-analysis.
The *VKORC1* variant, rs9923231, achieved *p-values* of 2E-3 in the African tract and 4E-3 in the meta-analysis.

Am J Hum Genet. 2025 Nov 6;112(11):2708-2719

Local Ancestry and Model Complexity

MODULE 4 · COMMIT FIRST

At this locus the effect is the same on every ancestry background and the allele frequencies are similar. In an admixed study of 500 people adjusted for age, sex and ten global PCs, which of global ancestry PCs alone, adding a local-ancestry covariate, and the local-ancestry interaction model detect it at p < 0.05?

A. All three

B. Global PCs and the local-ancestry covariate, but not the interaction model

C. Global ancestry PCs only

D. None of them

Local Ancestry and Model Complexity: the answer

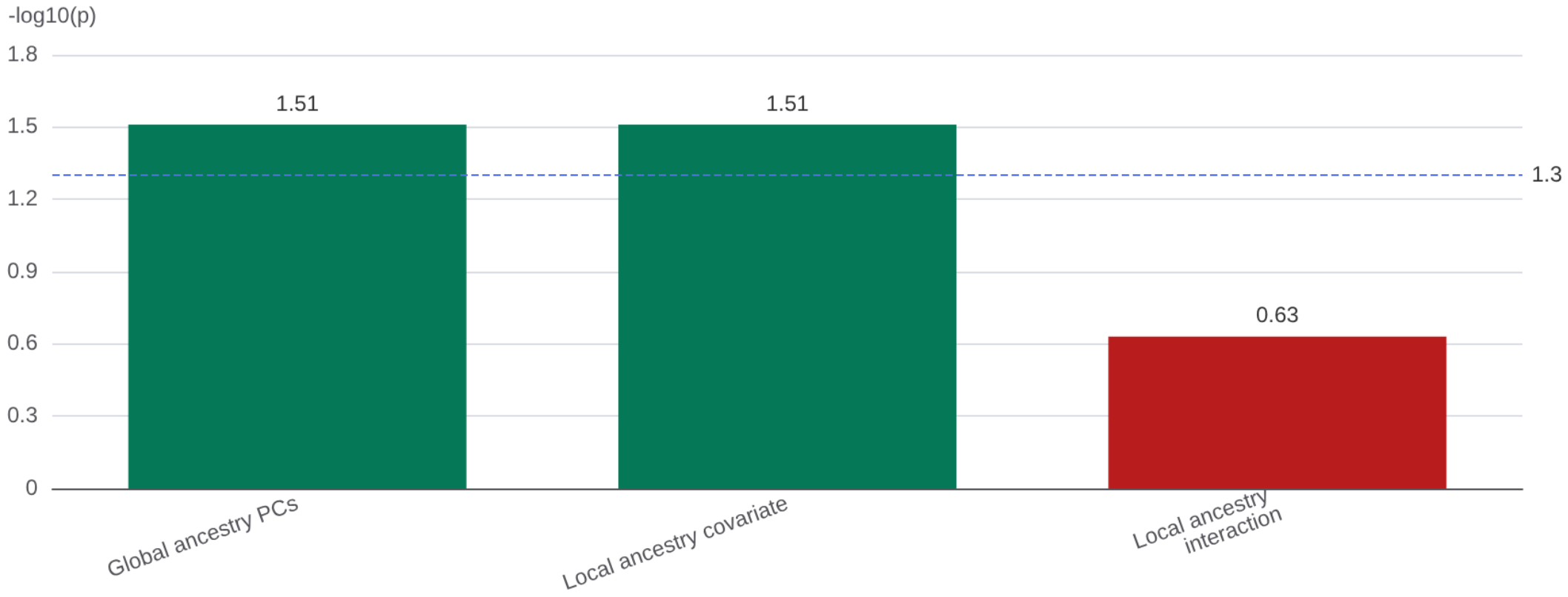
MODULE 4 · REVEAL

Answer

Global PCs and the local-ancestry covariate, but not the interaction model

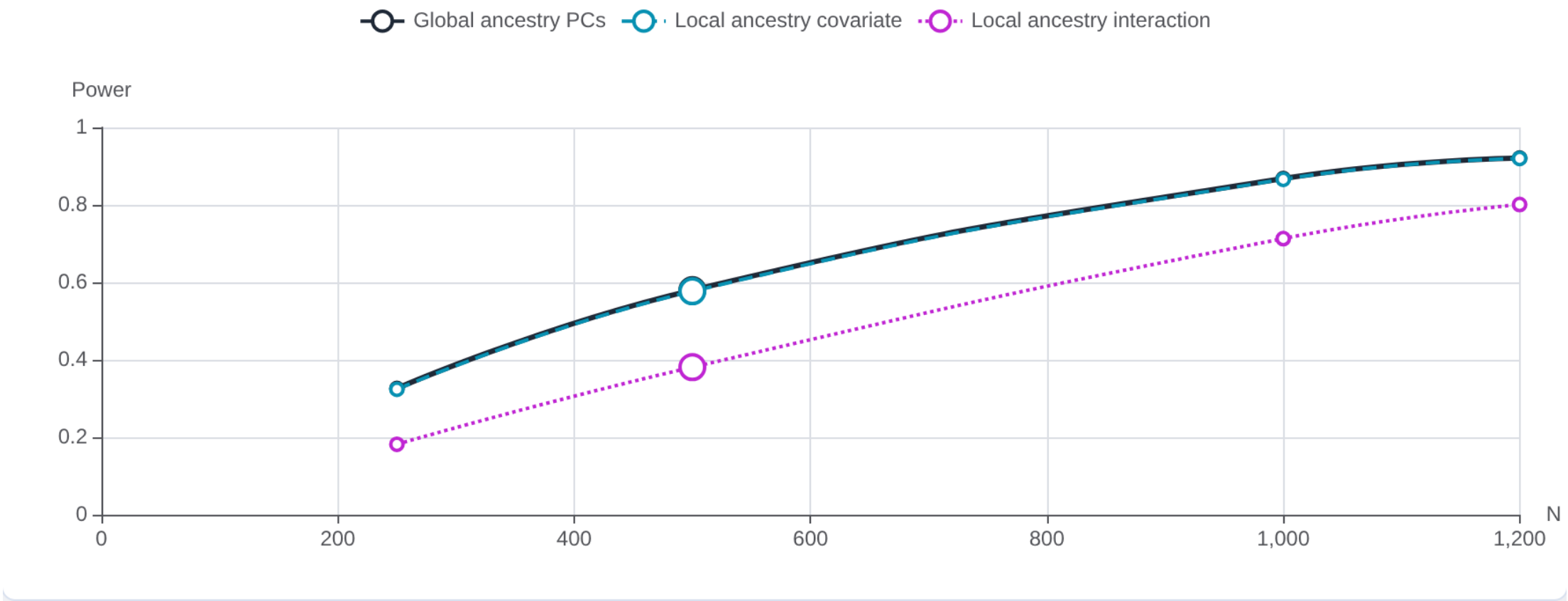
Module 4: Does each model detect the locus?

MODULE 4 · REVEAL



Module 4: Power by model and sample size

MODULE 4 · REVEAL



Polygenic Scores Across Ancestries

MODULE 5 · COMMIT FIRST

Three polygenic scores are trained on a cohort that is 59% European, 23% African and 18% Admixed American ancestry. At the starting shrinkage, which score is best calibrated in the smallest ancestry group?

A. The pooled-cohort PRS

B. The ancestry-aware PRS

C. The pooled PRS under strong shrinkage

D. They are equally calibrated

Polygenic Scores Across Ancestries: the answer

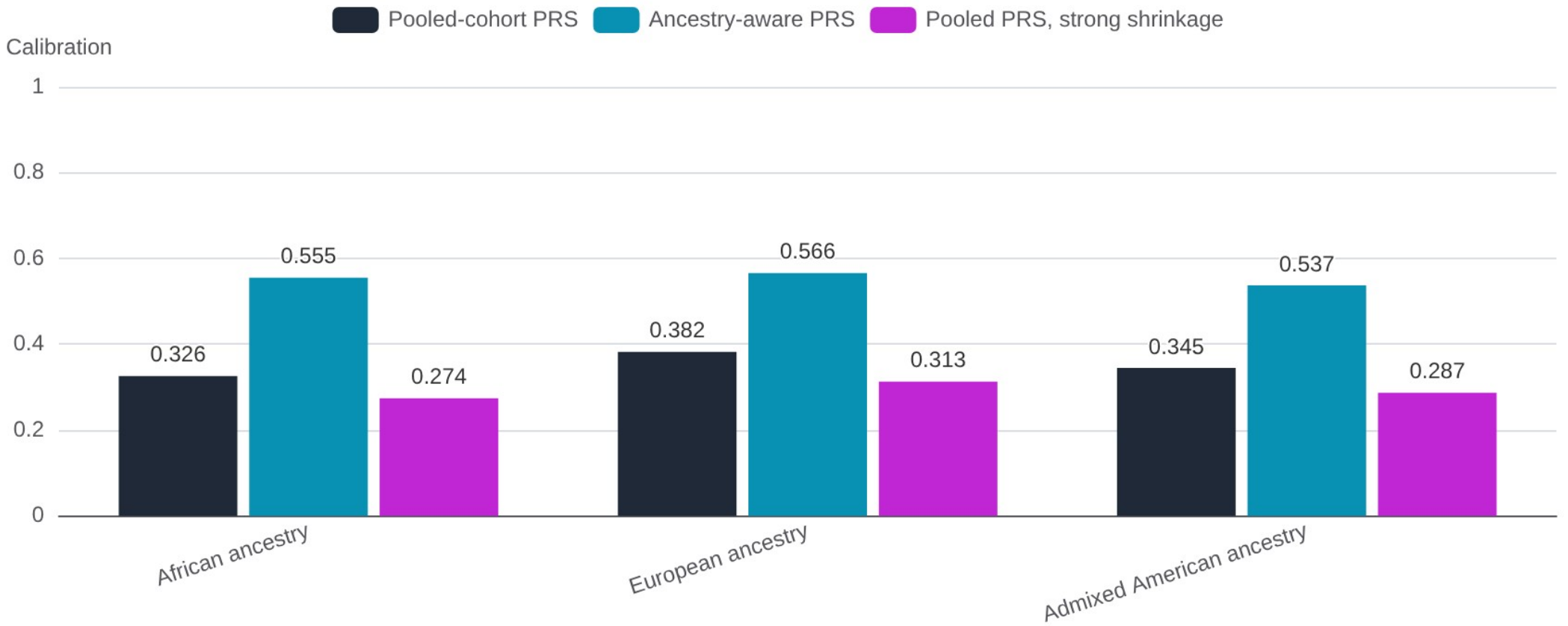
MODULE 5 · REVEAL

Answer

The ancestry-aware PRS

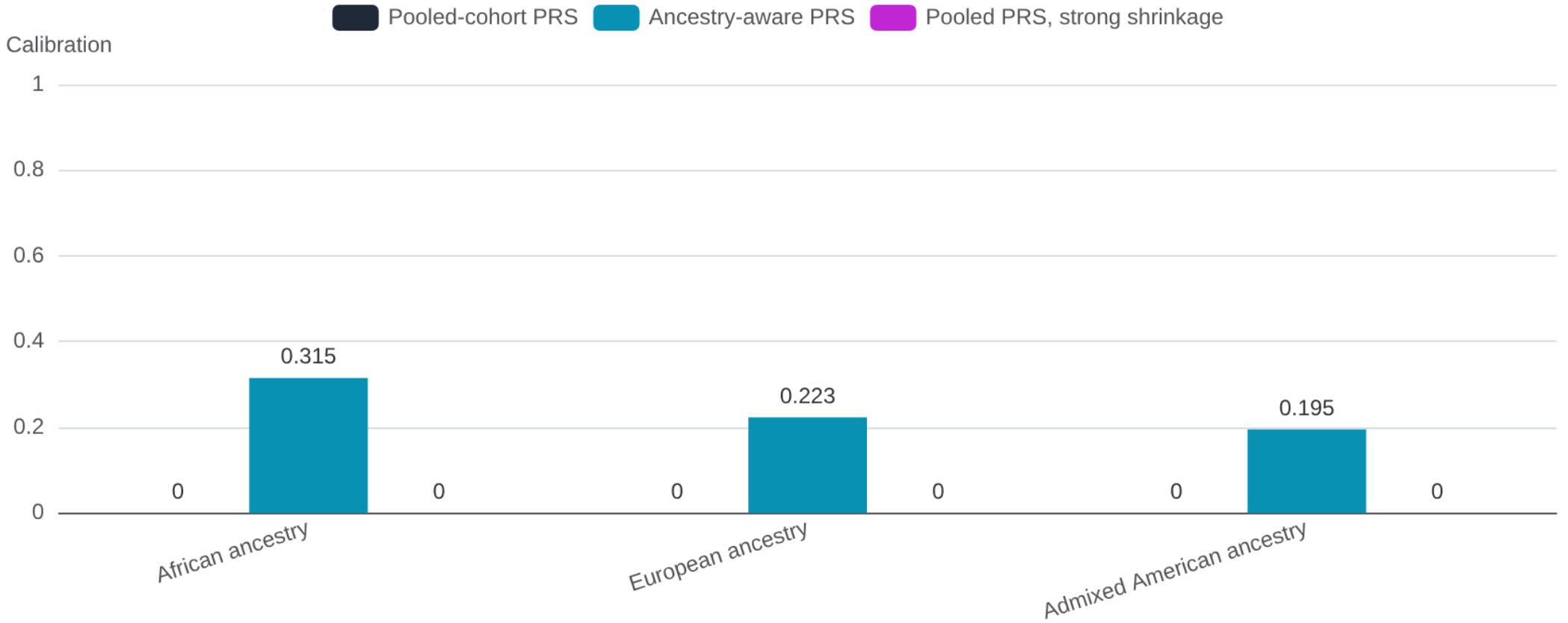
Module 5: Calibration at the starting shrinkage

MODULE 5 · REVEAL



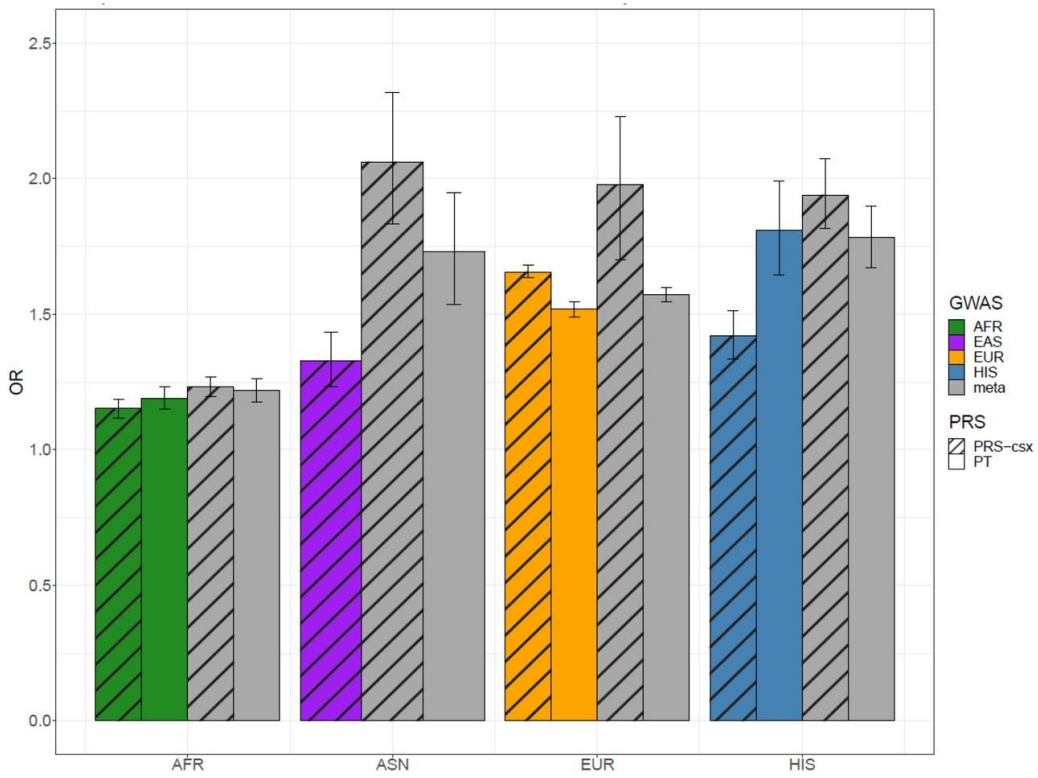
Module 5: Calibration at the strongest shrinkage

MODULE 5 · REVEAL



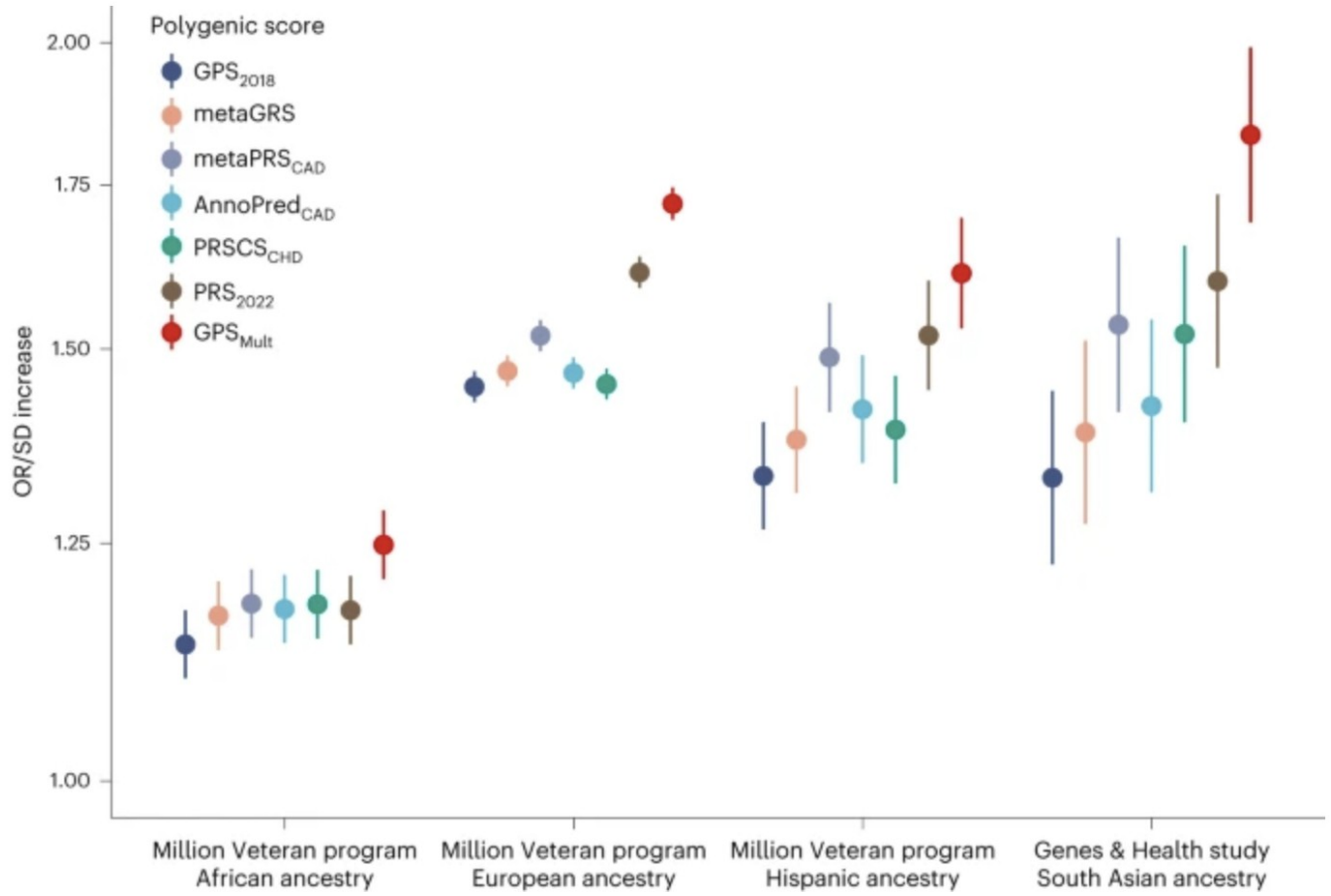
Polygenic Risk Scores: Cohorts used to inform PRS

- PRS models have historically been non-transferable between populations
- Several studies used a multi-ancestry GWAS to inform their PRS models to address this issue
- One PRS on CHD built from a cohort of 1.1 million people found that the multi-ancestry PRS models outperformed the ancestry-specific PRS models by a large margin for all groups except for AAs
- The multi-ancestry cohort used by this study was still heavily biased
    - Only 31K AFR patients (~3% of cohort) and not all were AAs
- Highlights the need for more studies comparing PRS models built on different cohorts for AAs.



Smith et al, MedRxiv (2024)

Fig. 4: External validation of GPS_{Mult} and benchmarking against published polygenic scores for CAD across multiple ancestries in Million Veteran Program and Genes & Health studies.



- PRS was created with trait-specific GWAS across ancestries.
- Score performed well across the entire cohort.
  - GPS_{Mult} was also associated with incident CAD events (hazard ratio per standard deviation 1.73, 95% confidence interval 1.70–1.76, $P < 0.001$)

Nature Medicine **volume 29**, pages 1793–1803 (2023)

Scalable Inference and Who Can Afford It

MODULE 6 · COMMIT FIRST

Three genome-wide scans are queued on a Bridges-2 RM node (128 cores) for 8 million variants in 250,000 people: a standard GWAS scan, a local ancestry-aware scan (4x the work), and a permutation scan (ancestry interaction with 1,000 permutations, 4,000x). Which finish within one week?

A. All three scans

B. The standard GWAS scan and the local ancestry-aware scan, but not the permutation scan

C. The standard GWAS scan only

D. None of them

Scalable Inference and Who Can Afford It: the answer

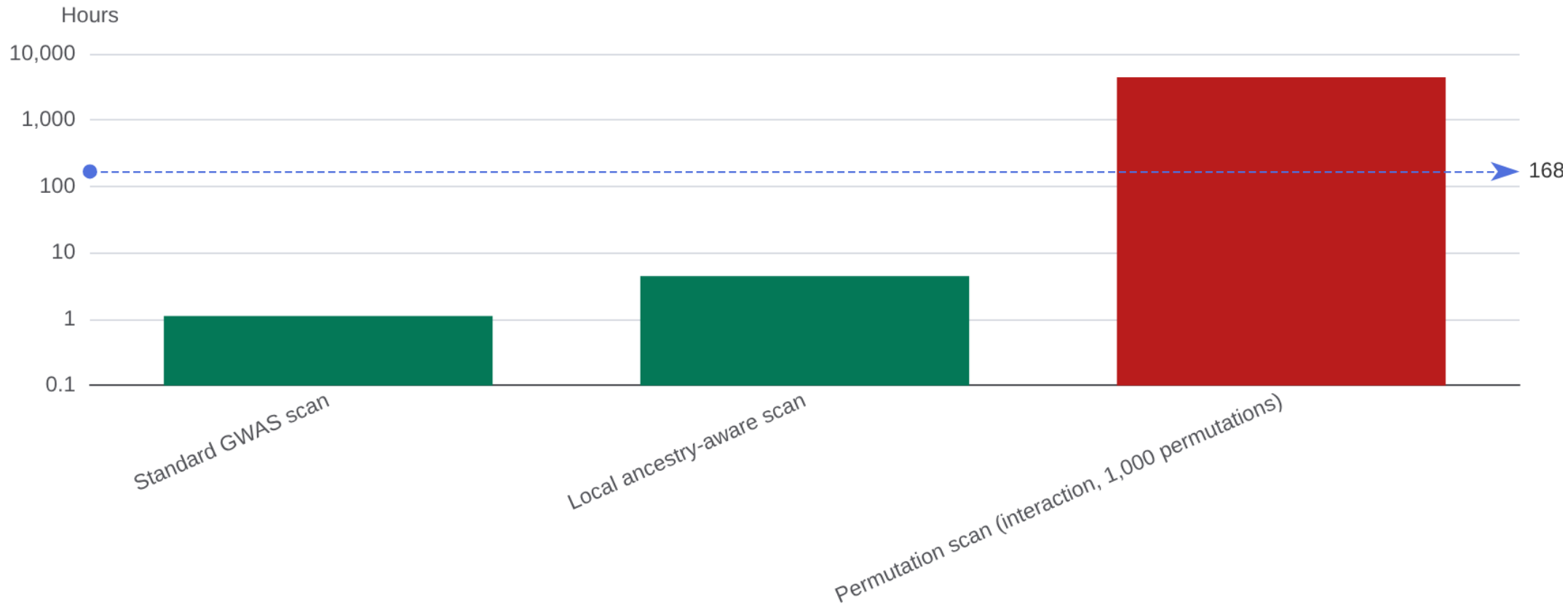
MODULE 6 · REVEAL

Answer

The standard GWAS scan and the local ancestry-aware scan, but not the permutation scan

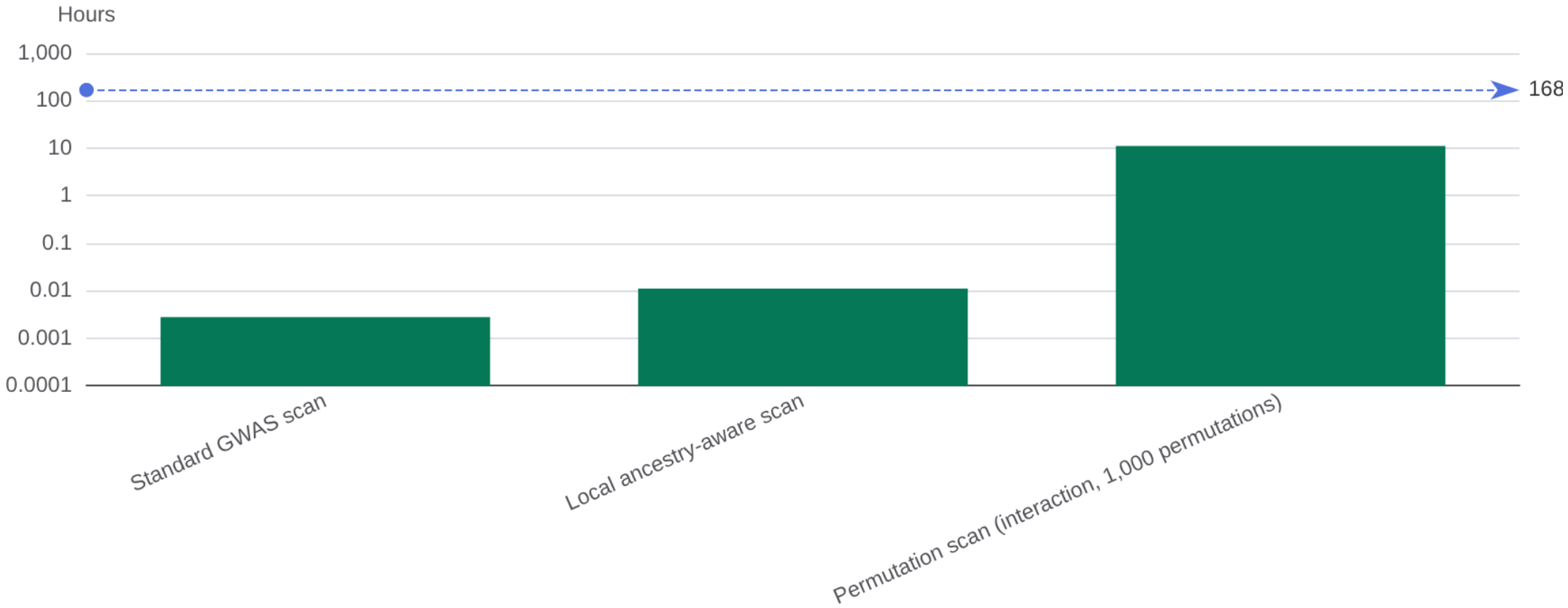
Module 6: One-week feasibility on one RM node

MODULE 6 · REVEAL



Module 6: One-week feasibility on one GPU node

MODULE 6 · REVEAL



Same code on CPU or GPU

MODULE 6 · RESOURCES

PyTorch one code path, CPU or GPU

JAX compiled for CPU, GPU or TPU

CuPy · cuDF NumPy and pandas on the GPU

Numba Python loops on threads or CUDA

Dask a laptop's cores to a cluster

Genetics tools that already scale

MODULE 6 · RESOURCES

PLINK 2 association and QC, multi-threaded

REGENIE whole-genome regression, multi-threaded

Hail Spark clusters; gnomAD, All of Us

tensorQTL QTL mapping on GPU (PyTorch)

Parabricks GPU alignment and variant calling

NSF ACCESS research HPC at no cost (Bridges-2)

What the data said: modules 1–3

SYNTHESIS · WHAT THE ROOM PREDICTED, WHAT THE DATA SAID

Module	The answer
1 · Pooled vs Stratified Analysis	It grows toward the strong European-ancestry effect
2 · Within-Ancestry Testing vs Meta-Analysis	It stays narrow and significant
3 · Global vs Local Ancestry Modeling	Local ancestry interaction

What the data said: modules 4–6

SYNTHESIS · WHAT THE ROOM PREDICTED, WHAT THE DATA SAID

Module	The answer
4 · Local Ancestry and Model Complexity	Global PCs and the local-ancestry covariate, but not the interaction model
5 · Polygenic Scores Across Ancestries	The ancestry-aware PRS
6 · Scalable Inference and Who Can Afford It	The standard GWAS scan and the local ancestry-aware scan, but not the permutation scan

Thank you!

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HEART-GeN Lab
A CompBio Group

Northwestern Medicine®
Feinberg School of Medicine

NIH National Institute on Minority Health and Health Disparities